

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

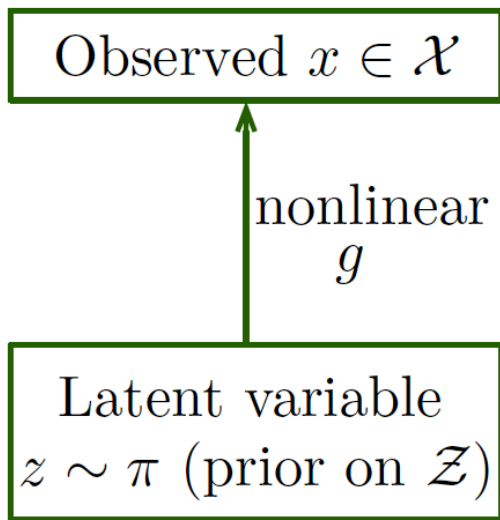
VARIATIONAL AUTOENCODERS FROM THE GROUND UP

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Latent Variable Model



- Let $\{p_\theta: \theta \in \Theta\}$ be a family of distributions on $\mathcal{X}$
- Learn nonlinear map $g: \mathcal{Z} \rightarrow \Theta$ (e.g., neural net)
- $\Pr(x, z|g) = \pi(z)p_{g(z)}(x)$

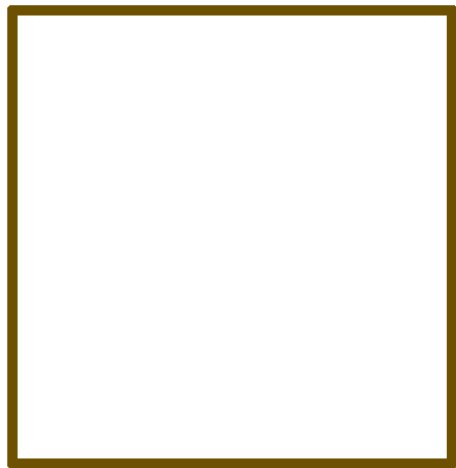
Example:

- $\mathcal{Z} = \mathbb{R}^k$ and π is Gaussian $N(0, I_k)$
- $\mathcal{X} = \{0,1\}^d$
- $\Theta = \{0,1\}^d$, p_θ is product-Bernoulli:

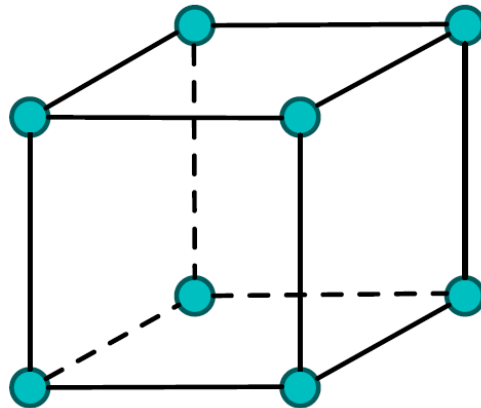
$$p_\theta(x) = \prod_{i=1}^d \theta_i^{x_i} (1 - \theta_i)^{1-x_i}$$

Example: Binarized MNIST

$$\mathcal{Z} = \mathbb{R}^k$$

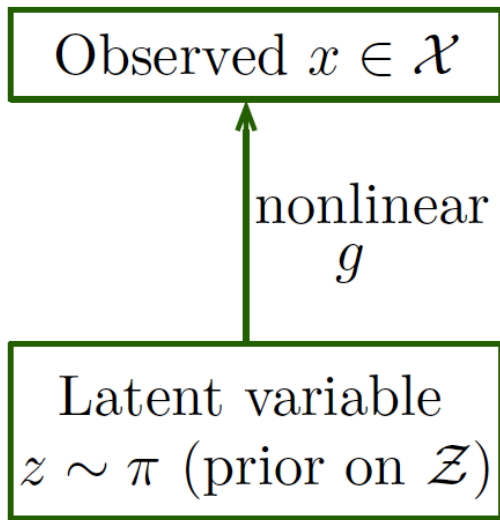


$$\mathcal{X} = \{0,1\}^{784}$$



- $g(z)$ is a product-Bernoulli distribution over $\{0,1\}^{784}$
- Why do we need a map f in the opposite direction?

Approximate Encoder



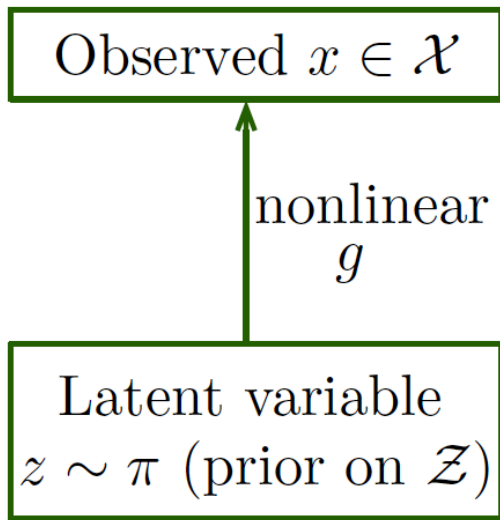
In order to learn g , need $\Pr(x|g)$: intractable!

$$\Pr(x|g) = \int_{\mathcal{Z}} \pi(z) p_{g(z)}(x) dz$$

Also intractable: $\Pr(z|x, g) = \Pr(x, z|g)/\Pr(x|g)$

- $\{p_{\theta}: \theta \in \Theta\}$, family of distributions on $\mathcal{X}$
- Neural net $g: \mathcal{Z} \rightarrow \Theta$
- $\Pr(x, z|g) = \pi(z) p_{g(z)}(x)$

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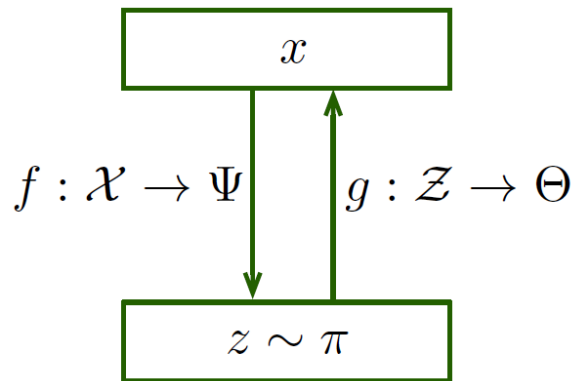
Learn another neural net to approximate $\Pr(z|x, g)$

- Let $\{q_\psi: \psi \in \Psi\}$ be a family of distributions on $\mathcal{Z}$
- Encoding map $f: \mathcal{X} \rightarrow \Psi$
- Want: $q_{f(x)}(z) \approx \Pr(z|x, g)$

E. g., $\mathcal{Z} = \mathbb{R}^k$, could take $\Psi = \mathbb{R}^{2k}$ with

- $\psi = (\mu_1, \dots, \mu_k, \ln \sigma_1, \dots, \ln \sigma_k)$
- $q_\psi = N\left(\mu, \text{diag}(\sigma_1^2, \dots, \sigma_k^2)\right)$

The Loss Function



- $\{p_\theta : \theta \in \Theta\}$, family of distributions on $\mathcal{X}$
- $\{q_\psi : \psi \in \Psi\}$, family of distributions on $\mathcal{Z}$
- $\Pr(x, z|g) = \pi(z)p_{g(z)}(x)$
- Want $q_{f(x)}(z) \approx \Pr(z|x, g)$

Should choose g to maximize

$$\Pr(x|g), \text{ which equals } \frac{\Pr(x, z|g)}{\Pr(z|x, g)}$$

Instead choose g, f to maximize

$$L_{f,g}(x) = \mathbb{E}_{z \sim q_{f(x)}} \ln \frac{\Pr(x, z|g)}{q_{f(x)}(z)}$$

This is called the ELBO.

Given data $x_1, \dots, x_n \in \mathcal{X}$, solve:

$$\arg \max_{f, g} \sum_{i=1}^n L_{f,g}(x_i)$$

Property of the ELBO

Claim: $L_{f,g}(x) = \ln \Pr(x|g) - K(q_{f(x)}, \Pr(\cdot | x, g))$

$$\begin{aligned} L_{f,g}(x) &= \mathbb{E}_{z \sim q_{f(x)}} \left[\ln \frac{\Pr(x, z|g)}{q_{f(x)}(z)} \right] \\ &= \mathbb{E}_{z \sim q_{f(x)}} \left[\ln \left(\frac{\Pr(x, z|g)}{q_{f(x)}(z)} \cdot \frac{\Pr(z|x, g)}{\Pr(z|x, g)} \right) \right] \\ &= \mathbb{E}_{z \sim q_{f(x)}} \left[\ln \left(\frac{\Pr(x, z|g)}{\Pr(z|x, g)} \cdot \frac{\Pr(z|x, g)}{q_{f(x)}(z)} \right) \right] \\ &= \mathbb{E}_{z \sim q_{f(x)}} \left[\ln \Pr(x|g) - \ln \frac{q_{f(x)}(z)}{\Pr(z|x, g)} \right] \\ &= \ln \Pr(x|g) - \mathbb{E}_{z \sim q_{f(x)}} \left[\ln \frac{q_{f(x)}(z)}{\Pr(z|x, g)} \right] \end{aligned}$$